

SyllabiX : AI-Powered Curriculum Generation Platform

Project Description

SyllabiX is an AI-powered Curriculum and Learning Intelligence Platform developed to automate curriculum design, academic planning, and learning resource generation. The platform leverages advanced Large Language Models, including IBM Granite and Groq Llama 3, to generate structured, outcome-based syllabi, personalized study plans, assessments, and academic handbooks. It is designed to support educators, academic institutions, corporate trainers, and self-directed learners by reducing the time and effort required for curriculum development.

The system integrates multiple educational functionalities into a single platform, including curriculum generation, quiz and assessment creation, personalized learning roadmaps, and AI-assisted academic support. A key component of the platform is DocMentor, an AI-powered PDF tutor that enables users to interact with educational documents through intelligent question-answering and content summarization. The platform also includes a gamified student workspace, performance analytics dashboards, and an AI Copilot that provides guidance and support throughout the learning process.

Developed using React, Flask, SQLite, and modern Generative AI technologies, SyllabiX enhances educational productivity, improves learning outcomes, and enables the rapid creation of industry-aligned academic programs. By combining curriculum intelligence, assessment automation, and personalized learning support, the platform provides a comprehensive solution for modern educational and training environments.

Scenarios

Scenario 1: University Curriculum Design

A university plans to introduce a new Bachelor's program in Artificial Intelligence and Machine Learning. Faculty members use SyllabiX to automatically generate semester-wise course structures, learning outcomes, and assessment plans. The platform suggests relevant textbooks, online resources, and practical projects aligned with industry standards. It ensures that topics are organized without gaps or unnecessary repetition. This significantly reduces curriculum development time and helps institutions launch new programs faster.

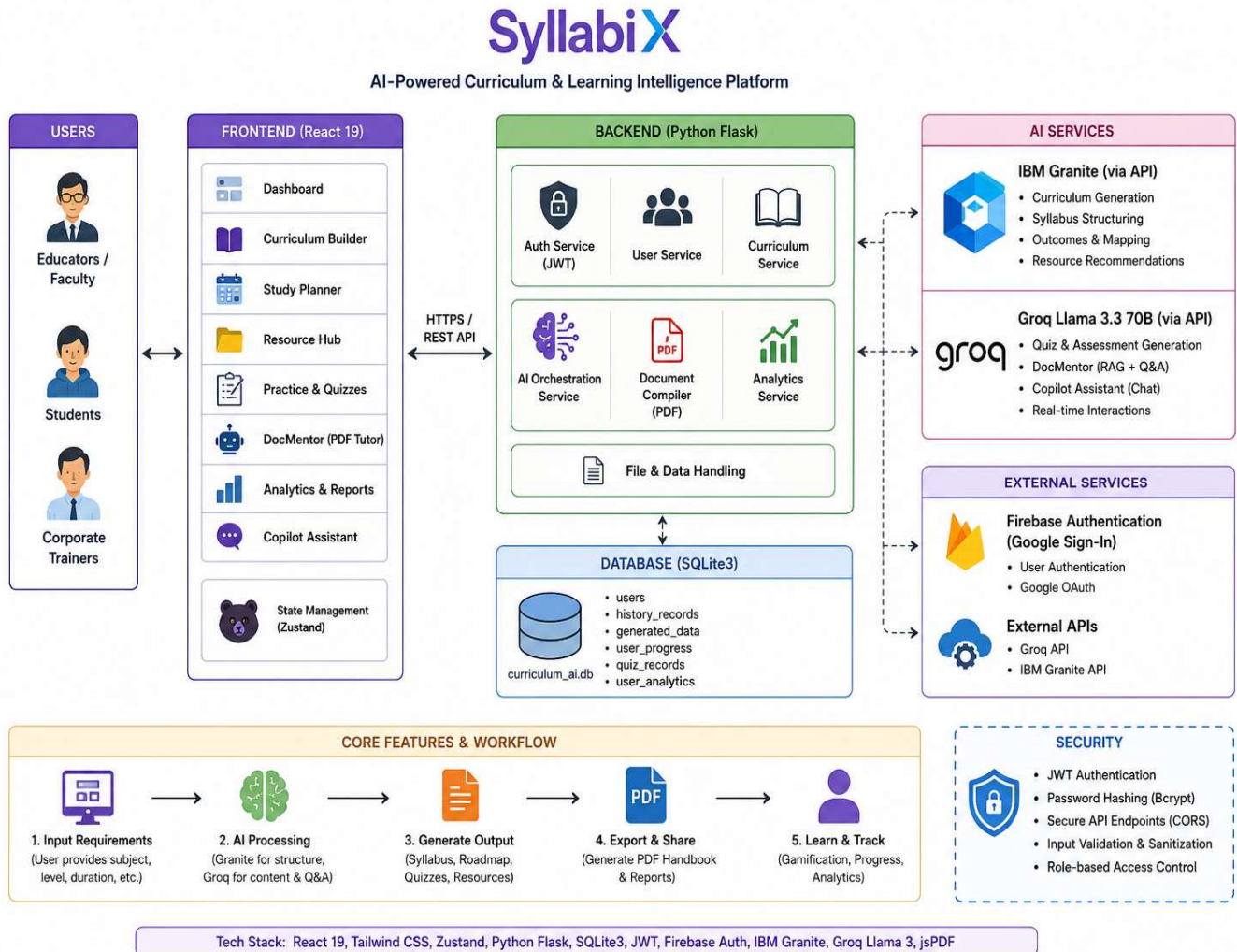
Scenario 2: Student Self-Learning and Career Preparation

A student preparing for a career in Data Science uses SyllabiX to create a personalized learning roadmap. Based on the selected domain and study duration, the platform generates a structured study plan with daily and weekly goals. It provides quizzes, recommended resources, and progress-tracking features to monitor learning. The AI-powered DocMentor helps answer questions from uploaded study materials. This enables students to learn efficiently and stay motivated through gamified achievements and badges.

Scenario 3: Corporate Training and Employee Upskilling

A technology company wants to train its employees in Cloud Computing and Generative AI. Training managers use SyllabiX to create customized learning paths tailored to organizational requirements. The platform automatically generates training modules, practice assessments, and study schedules for employees. Performance analytics and progress reports help managers evaluate learning outcomes. This ensures consistent skill development while reducing the effort required to design and manage training programs.

Architecture Overview



Pre-requisites and Setup

1. Node.js Installation: Install Node.js Version 18 or above (includes npm).

Reference: <https://nodejs.org/en/download>

2. Python Installation: Install Python Version 3.8 or above for running the Flask backend services.

Reference: <https://www.python.org/downloads/>

3. Backend Dependencies: Install all required Python packages inside the backend directory using: `pip install -r requirements.txt`

Reference: https://pip.pypa.io/en/stable/cli/pip_install/

4. Frontend Dependencies: Install all React and Vite dependencies inside the frontend directory using: `npm install`

Reference: <https://docs.npmjs.com/cli/v9/commands/npm-install>

5. Database Setup: SyllabiX uses SQLite for data storage. No separate database server installation is required. The database file is automatically created when the Flask application runs for the first time.

Reference: <https://www.sqlite.org/docs.html>

6. Backend Configuration: Create a `.env` file inside the backend directory and add the required configuration values:

`JWT_SECRET_KEY=your_secret_key`

`GROQ_API_KEY=your_groq_api_key`

`IBM_API_KEY=your_ibm_api_key`

`FLASK_ENV=development`

Reference: <https://pypi.org/project/python-dotenv/>

7. Frontend Configuration: Create a .env file in the frontend root directory and configure Firebase credentials:

VITE_FIREBASE_API_KEY=your_api_key

VITE_FIREBASE_AUTH_DOMAIN=your_auth_domain

VITE_FIREBASE_PROJECT_ID=your_project_id

VITE_FIREBASE_STORAGE_BUCKET=your_storage_bucket

VITE_FIREBASE_MESSAGING_SENDER_ID=your_sender_id

VITE_FIREBASE_APP_ID=your_app_id

References: <https://vitejs.dev/guide/> , <https://console.firebase.google.com/>

8. Backend Launch: Start the Flask backend server from the backend directory using:

python app.py

Reference: <https://flask.palletsprojects.com/>

9. Frontend Launch: Start the React development server from the frontend directory using:

npm run dev

Reference: <https://docs.npmjs.com/cli/v9/commands/npm-run-script>

10. Verification: After launching both servers, verify the application using:

- **Vite Frontend Dev Server:** running at <http://localhost:5173/>
- **Flask Backend Server:** running at <http://localhost:5000/> (API)

Confirm that user registration, login, Google authentication, curriculum generation, quiz generation, PDF export, and history management functionalities are working correctly.

Project Work Flow

Milestone 1: Environment Setup and Project Initialization

Activity 1.1: Configure React, Flask, SQLite Database Settings, and Backend Initialization.

Activity 1.2: Configure environment variables, Firebase credentials, JWT secrets, Groq API key, and IBM Granite API key.

Activity 1.3: Install all required backend and frontend dependencies using pip and npm.

Activity 1.4: Run and verify backend and frontend development servers.

Milestone 2: Backend API Development using Flask

Activity 2.1: Develop JWT-based authentication and user management APIs.

Activity 2.2: Create modular APIs for curriculum generation, assessments, study planning, and history management.

Activity 2.3: Implement SQLite database models for platform data management.

Activity 2.4: Develop service and controller logic for curriculum generation, AI integration, quizzes, exports, and analytics.

Milestone 3: AI Integration with IBM Granite and Groq Llama 3.3

Activity 3.1: Integrate IBM Granite for AI-powered curriculum and syllabus generation.

Activity 3.2: Integrate Groq Llama 3.3 for quiz generation, evaluation, DocMentor, and Copilot services.

Activity 3.3: Enable PDF handbook generation and document export functionality using jsPDF.

Milestone 4: React Frontend Development

Activity 4.1: Build responsive dashboards with curriculum insights, planners, assessments, and AI tools.

Activity 4.2: Develop curriculum generation, study planner, quiz, profile, and administration pages.

Milestone 5: Academic Curriculum Generator Module

Activity 5.1: Develop a multi-step curriculum generation interface for educators and learners.

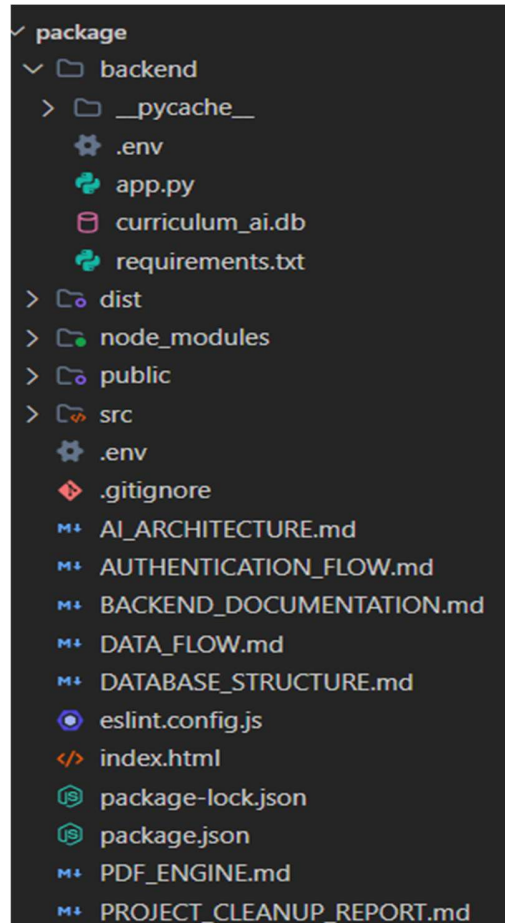
Activity 5.2: Implement semester-wise and subject-wise curriculum viewing and export features.

Activity 5.3: Build curriculum history management, retrieval, and user progress tracking functionality.

Milestone 1: Environment Setup and Project Initialization

Description

This milestone focuses on establishing the development environment required for the SyllabiX platform. The project structure is organized into separate frontend and backend modules to maintain a clean architecture. React and Flask frameworks are configured along with SQLite database connectivity. Environment variables are set up to securely manage API keys and authentication credentials. All required dependencies are installed, and both development servers are tested to ensure successful communication between system components. This milestone lays the foundation for all future development activities.




```
PS C:\Users\daksh\curriculum-ai> npm --version
10.9.0
PS C:\Users\daksh\curriculum-ai> node -v
v22.12.0
PS C:\Users\daksh\curriculum-ai> npm run dev

> curriculum-ai@0.0.0 dev
> vite

11:50:08 am [vite] (client) Re-optimizing dependencies because vite config has changed
Port 5173 is in use, trying another one...

VITE v8.0.16 ready in 579 ms

→ Local:   http://localhost:5174/
→ Network: use --host to expose
→ press h + enter to show help
```

Activity:1.1 Configure React, Flask, SQLite database settings, and backend initialization.

In this activity, the frontend is configured using React and Vite to provide a modern and responsive user interface. The backend is initialized using Flask and connected with SQLite for data storage. Database configurations and application settings are established to support future modules. Essential routing and server initialization procedures are implemented. This setup ensures smooth communication between the frontend, backend, and database layers.

```
vite.config.js > ...
1  import { defineConfig } from 'vite'
2  import react from '@vitejs/plugin-react'
3  import tailwindcss from '@tailwindcss/vite'
4
5  export default defineConfig({
6    | plugins: [react(), tailwindcss()], → Tab to Jump
7  })
8
```



```
app.py ×
backend > app.py
1  import os
2  import sqlite3
3  import bcrypt
4  from datetime import timedelta
5  from flask import Flask, request, jsonify, g
6  from flask_cors import CORS
7  from flask_jwt_extended import (
8      JWTManager, create_access_token, jwt_required,
9      get_jwt_identity, create_refresh_token
10 )
11 from dotenv import load_dotenv
12
13 load_dotenv()
14
15 app = Flask(__name__)
16
```

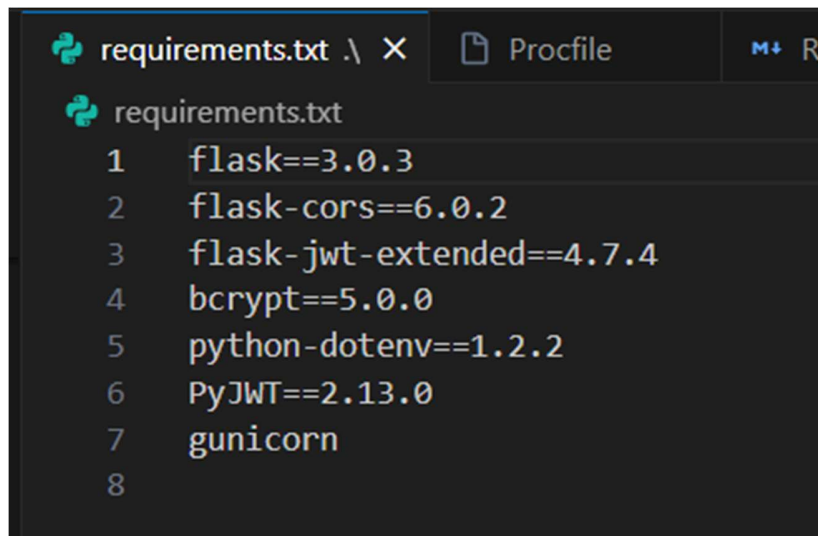
Activity 1.2: Configure environment variables, Firebase credentials, JWT secrets, Groq API key, and IBM Granite API key.

Environment configuration files are created to securely store API keys and application secrets. Firebase credentials are configured to support authentication services. JWT secret keys are added to enable secure user authentication and authorization. Groq and IBM Granite API keys are configured for AI-powered curriculum and assessment generation. This approach improves security and simplifies deployment management.

```
3
4  const GROQ_API_URL = "https://api.groq.com/openai/v1/chat/completions";
5
6  // Retrieve API key from Vite environment
7  const getApiKey = (): string => {
8      |   return import.meta.env.VITE_GROQ_API_KEY || '';
9  };
10
```

Activity 1.3: Install all required backend and frontend dependencies using pip and npm.

All necessary frontend and backend dependencies are installed using npm and pip package managers. React libraries, Firebase SDKs, Tailwind CSS, and UI components are configured on the frontend. Flask libraries, JWT authentication packages, SQLite utilities, and AI integration modules are installed on the backend. Dependency management ensures compatibility across project modules. This activity prepares the development environment for implementation.



```
requirements.txt \ X Procfile M+ R
requirements.txt
1 flask==3.0.3
2 flask-cors==6.0.2
3 flask-jwt-extended==4.7.4
4 bcrypt==5.0.0
5 python-dotenv==1.2.2
6 PyJWT==2.13.0
7 gunicorn
8
```

```
PS C:\Users\daksh\curriculum-ai> pip install -r requirements.txt
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: flask==3.0.3 in C:\Users\daksh\AppData\Roaming\Python\Python313\site-packages (from -r requirements.txt (line 1)) (3.0.3)
Requirement already satisfied: flask-cors==6.0.2 in C:\Users\daksh\AppData\Roaming\Python\Python313\site-packages (from -r requirements.txt (line 2)) (6.0.2)
Requirement already satisfied: flask-jwt-extended==4.7.4 in C:\Users\daksh\AppData\Roaming\Python\Python313\site-packages (from -r requirements.txt (line 3)) (4.7.4)
Requirement already satisfied: bcrypt==5.0.0 in C:\Users\daksh\AppData\Roaming\Python\Python313\site-packages (from -r requirements.txt (line 4)) (5.0.0)
Requirement already satisfied: python-dotenv==1.2.2 in C:\Users\daksh\AppData\Roaming\Python\Python313\site-packages (from -r requirements.txt (line 5)) (1.2.2)
Requirement already satisfied: PyJWT==2.13.0 in C:\Users\daksh\AppData\Roaming\Python\Python313\site-packages (from -r requirements.txt (line 6)) (2.13.0)
Requirement already satisfied: gunicorn in C:\Users\daksh\AppData\Roaming\Python\Python313\site-packages (from -r requirements.txt (line 7)) (26.0.0)
Requirement already satisfied: Werkzeug>=3.0.0 in C:\Users\daksh\AppData\Roaming\Python\Python313\site-packages (from flask==3.0.3->-r requirements.txt (line 1)) (3.0.3)
Requirement already satisfied: Jinja2>=3.1.2 in C:\Users\daksh\AppData\Roaming\Python\Python313\site-packages (from flask==3.0.3->-r requirements.txt (line 1)) (3.1.6)
```

Activity 1.4: Run and verify backend and frontend development servers.

The Flask backend server and React frontend server are launched in development mode. Connectivity between the frontend and backend is tested through API requests and responses. Database initialization and authentication functionality are verified. AI service configurations are checked to ensure successful integration. Successful execution confirms that the development environment is ready for feature development.

Both the frontend and backend servers are already running in the background:

- **Vite Frontend Dev Server:** running at <http://localhost:5173/>
- **Flask Backend Server:** running at <http://localhost:5000/> (API routes registered at `/api/*`)

Milestone 2: Backend API Development using Flask

Description

This milestone establishes the backend foundation of the SyllabiX platform through a secure and scalable API architecture. User authentication, curriculum management, assessments, and history tracking functionalities are implemented using Flask. SQLite database models and service layers are developed to support platform operations. The backend serves as the communication layer between users, AI services, and stored data.

Activity 2.1: Develop JWT-based authentication and user management APIs.

Secure authentication APIs are developed using JWT tokens and password encryption. User registration, login, logout, and profile management functionalities are implemented. Protected routes ensure that only authorized users can access platform resources. Firebase authentication is integrated to support Google Sign-In functionality. This activity provides a secure identity management system for the platform.

backend >  app.py

```
107
108 import json
109
110 def init_db():
111     db_dir = os.path.dirname(DB_PATH)
112     if db_dir:
113         os.makedirs(db_dir, exist_ok=True)
114     with sqlite3.connect(DB_PATH) as conn:
115         conn.execute("""
116             CREATE TABLE IF NOT EXISTS users (
117                 id            INTEGER PRIMARY KEY AUTOINCREMENT,
118                 name          TEXT      NOT NULL,
119                 email         TEXT      NOT NULL UNIQUE,
120                 password      TEXT      NOT NULL,
121                 provider      TEXT      NOT NULL DEFAULT 'local',
122                 photo_url     TEXT,
123                 firebase_uid  TEXT,
124                 created_at    TEXT      NOT NULL DEFAULT (datetime('now')),
125                 reset_token   TEXT,
126                 phone         TEXT,
```

```
@app.route("/api/register", methods=["POST"])
def register():
    data = request.get_json(silent=True) or {}
    name     = (data.get("name") or "").strip()
    email    = (data.get("email") or "").strip().lower()
    password = data.get("password") or ""

    if not name or not email or not password:
        return jsonify({"error": "All fields are required"}), 400

    if len(password) < 8:
        return jsonify({"error": "Password must be at least 8 characters"}),

    db = get_db()
    existing = db.execute("SELECT id FROM users WHERE email = ?", (email,)).f
    if existing:
        return jsonify({"error": "Email already registered"}), 409

    hashed = bcrypt.hashpw(password.encode(), bcrypt.gensalt()).decode()
    cur = db.execute(
        "INSERT INTO users (name, email, password) VALUES (?, ?, ?)",
        (name, email, hashed)
```



```
javascript

import { Navigate, useLocation } from 'react-router-dom'
import useAuthStore from '../store/authStore'

export default function ProtectedRoute({ children }) {
  const isAuthenticated = useAuthStore(s =>
s.isAuthenticated())
  const location = useLocation()

  if (!isAuthenticated) {
    return <Navigate to="/login" state={{ from: location }}
replace />
  }

  return children
}
```

Activity 2.2: Create modular APIs for curriculum generation, assessments, study planning, and history management.

RESTful APIs are developed to handle curriculum generation, quizzes, study plans, and user history records. Each module follows a structured routing mechanism for maintainability and scalability. User requests are validated before processing. Standardized API responses are implemented for consistent communication. This modular architecture simplifies future enhancements and maintenance.

```
# — History Routes inside app.py

@app.route("/api/history", methods=["GET"])
@jwt_required()
def get_history():
    user_id = get_jwt_identity()
    role = request.args.get("role", "").strip()
    db = get_db()

    if role:
        cursor = db.execute(
            "SELECT * FROM history_records WHERE user_id = ?
AND role = ? ORDER BY date_generated DESC",
            (user_id, role)
        )
    else:
        cursor = db.execute(
            "SELECT * FROM history_records WHERE user_id = ?
ORDER BY date_generated DESC",
            (user_id,)
        )
```

```
// — Curriculum Outliner inside utils.js
```

```
export function generateCurriculum(form) {
  const { skill, level, semesters, weeklyHours,
    industryFocus, curriculumType } = form
  const semCount = Math.min(8, Math.max(1,
    parseInt(semesters) || 4))

  // Progressive list of core topics based on requested
  // domain (e.g. Machine Learning, Web Dev, etc.)
  const activeProgression = domainProgression[skill] ||
    getFallbackProgression(skill)
  const totalUnits = semCount * 5

  const semesterList = Array.from({ length: semCount }, (_,
    i) => {
    const semNumber = i + 1
    const bloomsLevels = ['Remember', 'Understand', 'Apply',
      'Analyze', 'Evaluate', 'Create']

    // Construct 5 Progressive Units for this Semester
    const units = Array.from({ length: 5 }, (_, uIdx) => {
      const k = (semNumber - 1) * 5 + uIdx
      const poolIndex = totalUnits > 1 ? Math.floor((k /
        (totalUnits - 1)) * 39) : 0
```

```
// — AI Quiz Generation & Evaluation in quizGroqService.ts
```

```
export async function generateQuizFromGroq(
  curriculum: any,
  count: number,
  difficulty: string,
  types: string[],
  quizMode: string = "Practice Quiz",
  examStyle: boolean = false
): Promise<QuizQuestion[]> {
  try {
    const prompt = `You are a professional educational
    assessment engine. Generate exactly ${count} quiz questions
    based on the curriculum.
    Course Name: ${curriculum.program_name} | Difficulty:
    ${difficulty} | Mode: ${quizMode} | Exam Style: ${examStyle}
    Syllabus topics to cover:
    ${JSON.stringify(curriculum.semesters)}
    Allowed Types: ${JSON.stringify(types)}
    Return JSON format: { "questions": [...] }`;
```

Database tables are designed to store user information, generated curricula, study plans, and activity history. Relationships between records are established to maintain data consistency. CRUD operations are implemented for efficient data management. SQLite provides lightweight and reliable storage without requiring additional database servers. This activity forms the persistent data layer of the platform

[illegible]

Business logic is organized into service and controller modules to improve maintainability. Controllers process API requests and coordinate communication between the database and AI services. Export generation, analytics processing, and quiz management functionalities are implemented. Proper error handling and validation mechanisms are added. This architecture enables efficient execution of backend operations.


```
javascript

// — PDF Document Semantic Text Extractor

export const extractTextFromPdf = async (arrayBuffer,
onProgress) => {
  const pdfjs = await loadPdfJS();
  const loadingTask = pdfjs.getDocument({ data: arrayBuffer
});
  const pdf = await loadingTask.promise;
  const numPages = pdf.numPages;
  const pageTexts = [];
  let fullText = "";

  for (let i = 1; i <= numPages; i++) {
    const page = await pdf.getPage(i);
    const textContent = await page.getTextContent();
    const pageText = textContent.items.map(item =>
item.str).join(" ");
    pageTexts.push({ pageNum: i, text: pageText });
    fullText += pageText + "\n";
    if (onProgress) onProgress(Math.round((i / numPages) *
100));
  }

  // Create semantic chunks for RAG processing
}

// — Curriculum Generation Controller

export function generateCurriculum(form) {
  const { skill, level, semesters, weeklyHours,
industryFocus, curriculumType } = form
  const semCount = Math.min(8, Math.max(1,
parseInt(semesters) || 4))
  const activeProgression = domainProgression[skill] ||
getFallbackProgression(skill)
  const totalUnits = semCount * 5

  const semesterList = Array.from({ length: semCount }, (_,
i) => {
    const semNumber = i + 1
    const bloomsLevels = ['Remember', 'Understand', 'Apply',
'Analyze', 'Evaluate', 'Create']

    const units = Array.from({ length: 5 }, (_, uIdx) => {
      const k = (semNumber - 1) * 5 + uIdx
      const poolIndex = totalUnits > 1 ? Math.floor((k /
(totalUnits - 1)) * 39) : 0
```

Milestone 3: AI Integration with IBM Granite and Groq Llama 3.3

Description

This milestone integrates artificial intelligence capabilities into the SyllabiX platform. IBM Granite and Groq Llama 3.3 models are connected to generate educational content, assessments, and learning assistance. AI services transform user inputs into structured academic resources. This milestone provides the intelligent core that powers curriculum generation and personalized learning experiences.

Activity 3.1: Integrate IBM Granite for AI-powered curriculum and syllabus generation.

IBM Granite is integrated to generate structured curricula, learning outcomes, and semester plans. User requirements are processed and transformed into organized academic content. The generated curriculum follows educational standards and learning objectives. Resource recommendations and subject sequencing are also provided. This activity automates the curriculum design process.

```
// Requests JSON quiz questions from Groq
export async function generateQuizFromGroq(
  curriculum: any,
  count: number,
  difficulty: string,
  types: string[],
  quizMode: string = "Practice Quiz",
  examStyle: boolean = false
): Promise<QuizQuestion[]> {
  const apiKey = import.meta.env.VITE_GROQ_API_KEY || '';
  const res = await
fetch("https://api.groq.com/openai/v1/chat/completions", {
  method: "POST",
  headers: {
    "Content-Type": "application/json",
    Authorization: `Bearer ${apiKey}`,
  },
},
```

Activity 3.2: Integrate Groq Llama 3.3 for quiz generation, evaluation, DocMentor, and Copilot services.

Groq Llama 3.3 is integrated to support quizzes, evaluations, and AI tutoring features. The model generates practice questions and learning assistance in real time. DocMentor enables users to interact with uploaded educational materials. The Copilot assistant provides guidance throughout the platform. This activity enhances user engagement through intelligent interactions.

```
// Query DocMentor with dynamic RAG context injection
export const queryDocMentor = async (query, chunks,
chatHistory = [], actionType = "") => {
  const relevantChunks = retrieveContext(query, chunks, 8);
  const contextStr = relevantChunks.map(c => `[Excerpt Page
${c.pageNum}]: ${c.text}`).join("\n\n");

  const messages = [
    { role: "system", content: "You are DocMentor. Respond
based ONLY on the provided PDF excerpts and cite pages." },
    ...chatHistory.slice(-6).map(msg => ({
      role: msg.sender === 'user' ? 'user' : 'assistant',
      content: msg.text
    })),
    { role: "user", content: `Context
Excerpts:\n${contextStr}\n\nUser Prompt: ${query}` }
  ];
```

Activity 3.3: Enable PDF handbook generation and document export functionality using jsPDF.

Export functionality is developed using jsPDF to generate downloadable academic documents. Users can export curricula, study plans, and learning resources as PDF files. Proper formatting and document structure are maintained during export. Generated documents can be stored and shared for future use. This activity improves accessibility and usability of educational content.

```
// — Main Handbook Exporter

export const downloadCurriculumPDF = async (curriculum) => {
  const doc = new jsPDF({ orientation: 'portrait', unit:
'mm', format: 'a4' })
  drawPageBorder(doc)

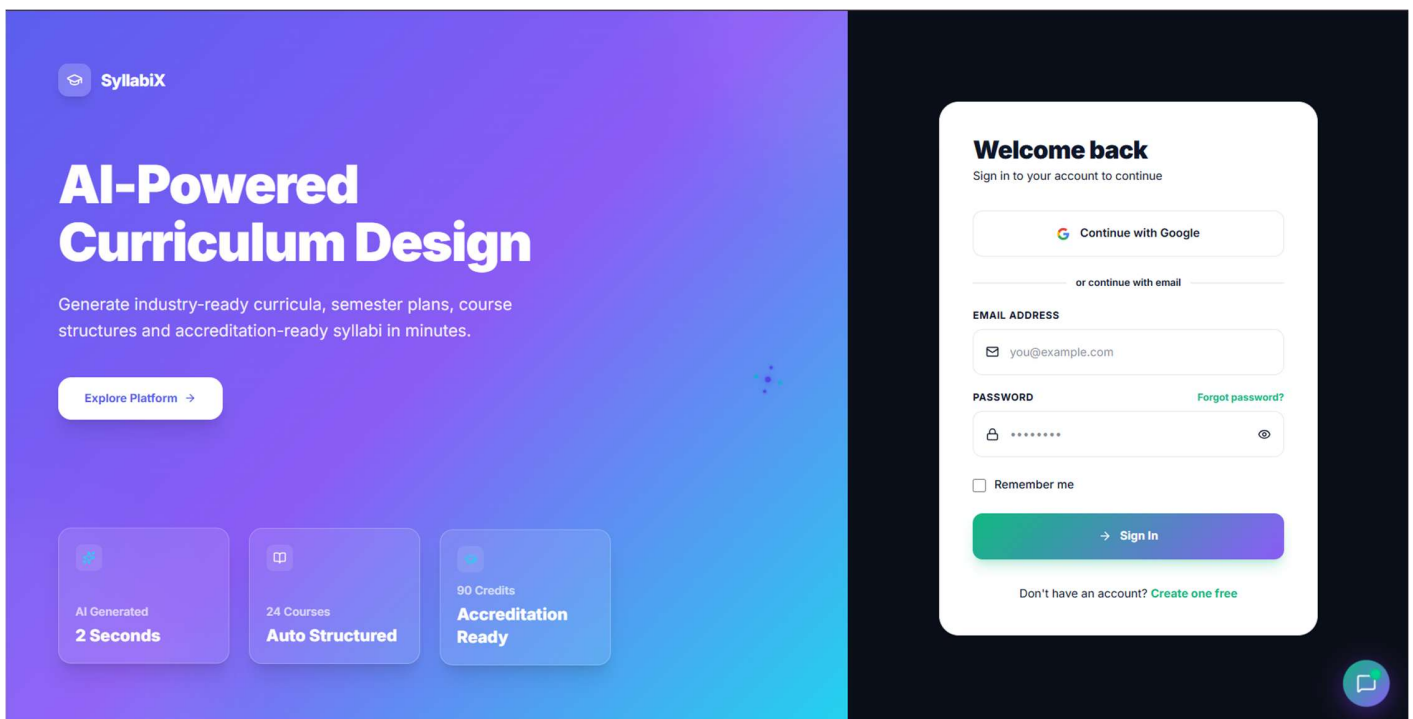
  // Cover Page specs Box
  const specBoxY = 98
  doc.setFillColor(248, 250, 252)
  doc.rect(15, specBoxY, 180, 58, 'F')
  doc.rect(15, specBoxY, 180, 58)

  const metaRows = [
    { label: 'Course Title', value: curriculum.program_name
},
    { label: 'Education Level', value:
curriculum.education_level },
    { label: 'Total Credits', value:
`${curriculum.stats?.totalCredits || 0} Credits` }
  ]
}
```

Milestone 4: React Frontend Development

Description

This milestone focuses on developing the user interface of the SyllabiX platform using React, Vite, Tailwind CSS, and modern frontend technologies. Responsive dashboards and interactive pages are created to provide an intuitive user experience. API integration mechanisms enable seamless communication with backend services. The frontend transforms complex backend operations into user-friendly workflows that support curriculum creation, assessments, learning analytics, and AI-powered educational assistance.



Activity 4.1: Build responsive dashboards with curriculum insights, planners, assessments, and AI tools.

A responsive dashboard is developed to provide users with quick access to curriculum generation, study planning, assessments, and AI-powered learning tools. Interactive cards and visual components display important information in an organized manner. Responsive layouts ensure compatibility across desktop and mobile devices. Navigation and user accessibility features are also implemented. This activity creates the central workspace for platform users.

Faculty & Student Dashboard

Choose Your Learning Role

Select how you would like to use the platform. Access tools tailored for teaching coordinators or personalized self-study.



ACADEMIC LEAD

Faculty Workspace

For Educators & Directors

Create complete academic curricula, generate unit-wise syllabus outlines, configure quizzes directly, and export PDF handbooks.

- ✓ Curriculum Generation with AI Outliner
- ✓ Bloom's Taxonomy Outcome Mapping
- ✓ Unit-specific Quiz Generator (any count)
- ✓ Reference Recommendations (Docs, YouTube, Papers)
- ✓ Accreditation Exports (PDF Handbook)



Enter Faculty Workspace >



ACTIVE LEARNER

Student Workspace

For Students & Learners

Generate custom study paths, take self-practice quizzes, plan your schedules, and earn achievements as you progress.

- ✓ Interactive Study Blueprints & Modules
- ✓ Self-Practice Quizzes with AI Feedback
- ✓ 30/60/90 Day Study Planner
- ✓ Skill Achievements & Cognitive Badges
- ✓ Targeted Resource Links (Docs, Books, Videos)

Enter Student Workspace >

FACULTY WORKSPACE

Academic outline & Syllabus designer

 **SyllabiX Builder**

 **Quiz Generator** 

 **History Logs**

 **Exports Manager**

STUDENT HUB

Self-study and exam hub

 **Student Dashboard**

 **Study Blueprint**

 **Syllabus Resources**

 **Practice Quizzes** 

 **Study Planner**

 **AI PDF Tutor**

 **Saved Roadmaps**

PROFESSIONAL CERTIFICATE PROGRAM

Professional Certificate in Machine Learning

8 Semesters · Aerospace Focus · 15 hrs/week

 Export PDF Regenerate**40**

Total Units

**120**

Total Credits

**160**

Topics Covered

**8**

Semesters

Program Objectives

- Master core Machine Learning concepts from foundations to advanced specialization.
- Build industry-ready projects aligned with Aerospace challenges.
- Apply Bloom's Taxonomy progression to refine technical learning outputs.
- Prepare for certification audits and software development requirements.

Target Career Roles

Senior Engineer Technical Architect

Domain Lead Research Specialist

Product Manager

Recommended Certifications

AWS Certified Developer

Google Professional Cloud Architect

CompTIA Security+



CAPSTONE PROJECT

AI-Powered Machine Learning Enterprise Pathway

End-to-end sandbox project simulating a real-world Aerospace platform deploy.

- Production Docker Blueprint
- Syllabus Alignment Audit
- Bloom's Rubric Checklist
- Working API Prototype
- GitHub CI Action workflow

Semester Roadmap Breakdown

S1

Semester 1

FOUNDATIONAL PRINCIPLES

Unit 1 Unit 2 Unit 3 Unit 4 Unit 5

5 Units • 15 Credits

S2

Semester 2

CORE & APPLIED SYSTEMS

Unit 1 Unit 2 Unit 3 Unit 4 Unit 5

5 Units • 15 Credits

S3

Semester 3

ADVANCED METHODOLOGIES

Unit 1 Unit 2 Unit 3 Unit 4 Unit 5

5 Units • 15 Credits

S4

Semester 4

INDUSTRY & SCALE DEPLOYMENT

Unit 1 Unit 2 Unit 3 Unit 4 Unit 5

5 Units • 15 Credits

Activity 4.2: Develop curriculum generation, study planner, quiz, profile, and administration pages.

Dedicated pages are designed to manage various platform functionalities. Curriculum generation pages collect user requirements and display AI-generated outputs. Study planner and quiz pages help users organize learning activities and assessments. Profile management pages allow users to update personal information and preferences. Administration pages provide additional control over platform resources and user management.

Profile Information



Upload profile photo

JPG, JPEG or PNG. Max size 2MB.

Choose File

Remove

FULL NAME

Dakshayani Mukku

EMAIL ADDRESS

dakshayanimukku2005@gmail.com

PHONE NUMBER

8019241252

INSTITUTION / ORGANIZATION

PVPSIT

ROLE

Student

COUNTRY

India

TIMEZONE

GMT+5:30 (IST)

BIO

Tell us about yourself

Cancel

Save Changes

Curriculum Designer

Select domains, target focus, and build unit-wise curriculum roadmaps.

SKILL / DOMAIN

Select Skill / Domain

EDUCATION LEVEL

Select Education Level

SEMESTERS

Select Number of Semesters

WEEKLY HOURS

Select Weekly Study Hours

INDUSTRY FOCUS

Select Industry Focus

CURRICULUM TYPE

Select Curriculum Type



Generate Academic Program Outline

Milestone 5: Academic Curriculum Generator Module

Description

This milestone delivers the core functionality of the SyllabiX platform by implementing the Academic Curriculum Generator. The module enables educators, institutions, and learners to generate customized academic curricula using artificial intelligence. Users can create, manage, export, and track curricula through an intuitive interface. This milestone transforms the platform into a complete educational planning and curriculum development solution.

Activity 5.1: Develop a multi-step curriculum generation interface for educators and learners.

A guided multi-step interface is designed to collect curriculum requirements from users. Inputs such as domain, program duration, educational level, industry focus, and learning objectives are captured systematically. Validation mechanisms ensure accurate and complete data entry. The interface simplifies the curriculum creation process for users with varying levels of expertise. This activity serves as the primary input mechanism for AI-powered curriculum generation.

Activity 5.2: Implement semester-wise and subject-wise curriculum viewing and export features.

Generated curricula are presented in structured semester-wise and subject-wise formats for easy understanding. Users can navigate through subjects, learning outcomes, and recommended resources. Export functionality allows curricula to be downloaded as professional PDF documents. Proper formatting and organization improve readability and usability. This activity enables users to access and distribute curriculum content efficiently.

Unit Roadmap Breakdown

Unit 1

Bloom: Remember

Unit 1: Python for Data Science Foundations

Comprehensive study of Python for Data Science Foundations focusing on practical integration, core properties, and performance parameters within Technology contexts.

SYLLABUS TOPICS COVERED

1 Python for Data Science Foundations

2 Configuring Python for Data Science Foundations components

3 Practical implementation of Python for Data Science Foundations layouts

4 Optimizing and debugging Python for Data Science Foundations codebases

UNIT LEARNING OUTCOMES

LO1.1: Understand the architecture and layout of Python for Data Science Foundations.

LO1.2: Apply theoretical parameters of Python for Data Science Foundations to build production-level software modules.

LO1.3: Evaluate the computational speed, security protocols, and memory footprints of Python for Data Science Foundations models.

RECOMMENDED STUDY MATERIALS

Learn Python for Data Science Foundations - YouTube Course

Python for Data Science Foundations Guide - Google Developers

Deep Dive into Python for Data Science Foundations - Research Pa...

python-for-data-science-foundations-starter

Required Textbook: Machine Learning Handbook: Practical Guide to Python for Data Science Foundations

PRACTICE ACTIVITIES

Set up local sandbox configurations for Python for Data Science Foundations.

Draft unit test suites validating Python for Data Science Foundations executions.

UNIT ASSESSMENTS

Diagnostic Multiple Choice Quiz for Python for Data Science Foundations.

Practical Lab Project 01 on Python for Data Science Foundations.

PERSONALIZED LEARNING PATH & STUDY GUIDE

STUDY BLUEPRINT

Structured Study Timeline & Learning Milestones

STUDY SPECIFICATIONS

STUDENT NAME

Dakshayani Mukku

STUDY STRATEGY

Biotechnology Applications

COURSE TITLE

Professional Certificate in Machine Learning

WEEKLY COMMITMENT

10 Hours/Week

EDUCATION LEVEL

Professional Certificate

STUDY DURATION

2 Semesters

DOMAIN/SKILL

Machine Learning

GENERATION DATE

6/11/2026

EXECUTIVE STUDY SUMMARY

This personalized study blueprint outlines the academic curriculum path designed for Dakshayani Mukku. By following the sequenced semesters, units, and day-wise study planners, you will build robust competency in the Machine Learning domain with a specific specialization in Biotechnology integrations. Each module lists targeted learning outcomes, recommended textbooks, and online resources to ensure absolute readiness for certifications and placements.

DOCUMENT METADATA

**Unit-2-A Text book (1).pdf**

61 pages · 29,800 words

IDENTIFIED CORE TOPICS

- ✔ Steganography
- ✔ Cyber Attacks
- ✔ Malware
- ✔ Network Security
- ✔ Password Cracking
- ✔ SQL Injection

DOCUMENT OUTLINE

- Introduction to Cyber Security
- Tools and Methods Used in Cybercrime
- Understanding Cyber Attacks
- Network Security and Vulnerabilities
- Types of Malware and Cyber Threats
- Cyber Forensics and Incident Response
- Legal Perspectives on Cyber Crime
- Computer Forensics and Cyber Security Measures

Welcome! I have successfully analyzed **Unit-2-A Text book (1).pdf**. I can tutor you on this document.

Here are some things we can do:

- Use the quick buttons below to generate custom study sheets.
- Ask questions about specific pages or sections.
- Test your comprehension on core topics.

Note: All answers are strictly retrieved from this document.

02:44 PM

Generate MCQs

Generate Short Answers

Generate Long Answers

Generate Viva Questions

Generate Interview Questions

Generate Revision Notes

Generate Summary

Generate Important Questions

Summarize the main points of this document.

What are the most difficult concepts explained here?

Explain the headings in the first chapter.

Ask DocMentor a question about the PDF contents...



Activity 5.3: Build curriculum history management, retrieval, and user progress tracking functionality.

Generated curricula are presented in structured semester-wise and subject-wise formats for easy understanding. Users can navigate through subjects, learning outcomes, and recommended resources. Export functionality allows curricula to be downloaded as professional PDF documents. Proper formatting and organization improve readability and usability. This activity enables users to access and distribute curriculum content efficiently.

PRACTICE PARAMETERS

QUESTION COUNT

5 Questions

DIFFICULTY

Medium

QUIZ MODE

Practice Quiz

☐ Exam Style Mode

QUESTION TYPES

☒ MCQ

☒ True/False

☒ Short Answer

☐ Long Answer

☒ Descriptive

☒ Scenario Based

☒ Application Based

☐ Case Study

☐ Interview Style

☒ Fill in the Blanks

Generate Practice Quiz

ACTIVE PRACTICE QUIZ (5 QUESTIONS)

Professional Certificate in Machine Learning

Curriculum-wide Quiz · Difficulty: Medium · Mode: Practice Quiz

Download Paper

Download Key

Cancel

- MCQ

What is the primary function of the self-attention mechanism in transformer architectures?

☐ A. To reduce the dimensionality of input data

☐ B. To increase the computational speed of the model

☐ C. To allow the model to attend to different parts of the input sequence simultaneously

☐ D. To improve the security protocols of the model
- TRUE/FALSE

True or False: Autoencoders are typically used for supervised learning tasks.

☐ True

Submit Answers

DAY-WISE STUDY SCHEDULE

Below is the customized milestone timeline generated for Dakshayani Mukku covering the course units. Complete the targets, review objectives, and execute practical labs accordingly.

Timeline	Syllabus Focus	Schedule Type	Focus & Practical Assignments
Day 1-1	Unit 1: Python for Data Science Foundations	Theory Study & Topics	Core study of: Python for Data Science Foundations, Configuring Python for Data Science Foundations components, Practical implementation of Python for Data Science Foundations layouts, Optimizing and debugging Python for Data Science Foundations codebases. Target Blooms tier: Remember.
Day 2	Unit 1: Python for Data Science Foundations	Practice Lab Day	Complete practice task: Set up local sandbox configurations for Python for Data Science Foundations.
Day 3	Unit 1: Python for Data Science Foundations	Revision & Quiz	Review notes and attempt: Diagnostic Multiple Choice Quiz for Python for Data Science Foundations.
Day 4	Unit 2: Exploratory Data Analysis & Visualizations	Study Focus	Study and map topics: Exploratory Data Analysis & Visualizations, Configuring Exploratory Data Analysis & Visualizations components. Check outcomes: LO2.1: Understand the architecture and layout of Exploratory Data Analysis & Visualizations.
Day 5	Unit 2: Exploratory Data Analysis & Visualizations	Practice & Quiz	Execute lab: Set up local sandbox configurations for Exploratory Data Analysis & Visualizations.. Take diagnostic unit assessment.
Day 6-6	Unit 3: Clustering Algorithms & K-Means	Theory Study & Topics	Core study of: Clustering Algorithms & K-Means, Configuring Clustering Algorithms & K-Means components, Practical implementation of Clustering Algorithms & K-Means layouts, Optimizing and debugging Clustering Algorithms & K-Means codebases. Target Blooms tier: Apply.

History Log

Browse, filter, and review previously generated academic curricula.

Q Search programs, skills, indi

All

Bachelor

Master

Diploma

Certificate

CURRICULUM NAME

DATE GENERATED

DOMAIN

ACTIONS

Diploma in Data Science

6/11/2026

Data Science

View

Delete

Diploma in Web Development

6/10/2026

Web Development

View

Delete

Certificate in Web Development

6/10/2026

Web Development

View

Delete

Diploma in Cloud Computing

6/9/2026

Cloud Computing

View

Delete

PhD in Cybersecurity

6/9/2026

Cybersecurity

View

Delete

Conclusion

SyllabiX successfully demonstrates the application of Artificial Intelligence in modern curriculum design and personalized learning. The platform combines React, Flask, SQLite, IBM Granite, and Groq Llama 3.3 to provide an intelligent environment for generating academic curricula, study plans, assessments, and learning resources. Through AI-powered automation, the system significantly reduces the time and effort required for curriculum development while ensuring structured and outcome-based educational content.

The platform also enhances the learning experience by providing interactive features such as quiz generation, DocMentor assistance, progress tracking, curriculum history management, and PDF export functionality. Secure authentication mechanisms and efficient database management ensure reliable operation and data security. By integrating advanced AI technologies with a user-friendly interface, SyllabiX serves as a comprehensive solution for educators, institutions, corporate trainers, and self-directed learners.

Overall, the project successfully achieves its objective of simplifying curriculum creation and improving educational planning through intelligent automation. SyllabiX establishes a strong foundation for future enhancements such as adaptive learning recommendations, collaborative curriculum design, advanced analytics, and integration with Learning Management Systems (LMS), making it a scalable and future-ready educational platform.